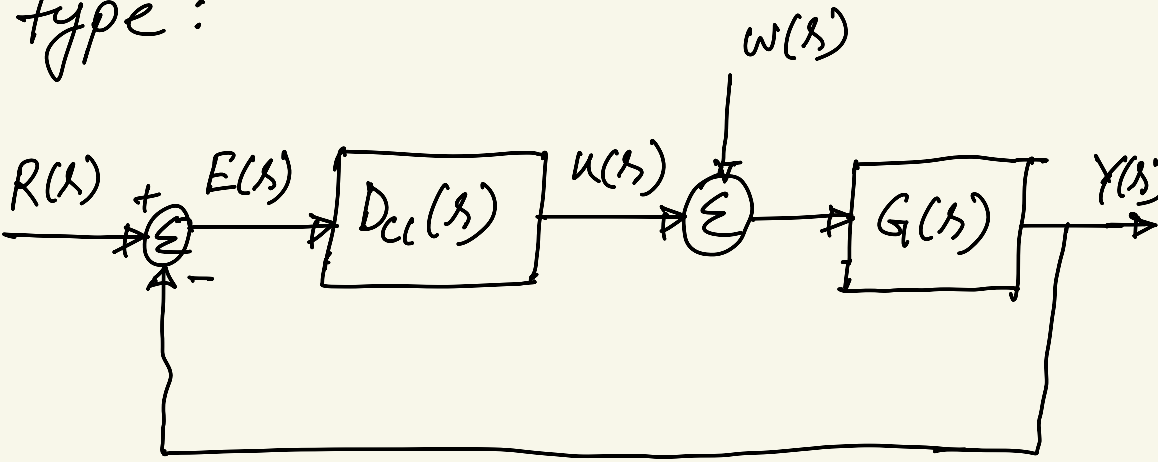
Steady-state error and system type:



Previously, we have derived that the response Y(s) of the closed loop system is as follows:

$$Y(s) = \frac{D_{cl}(s)\,G(s)}{1 + D_{cl}(s)\,G(s)} R(s) + \frac{G(s)}{1 + G(s)\,D_{cl}(s)} W(s)$$

$$= \frac{D_{cl}\,G}{1 + D_{cl}\,G} R + \frac{G}{1 + G D_{cl}} W$$

The closed-loop error is as follows

$$E(s) = R(s) - Y(s)$$

$$E = R - Y$$
$$= R - \left[\frac{D_{cl} G}{1 + D_{cl} G} \cdot R + \frac{G}{1 + G D_{cl}} \cdot W\right]$$

$$E = \frac{1}{1 + G D_{cl}} R - \frac{G}{1 + G D_{cl}} W$$

If $W = 0$

$$E = \frac{1}{1 + G D_{cl}} R$$

Steady-state error is the error when $t \to \infty$.

$$\lim_{t \to \infty} e(t) = e_{ss} = \lim_{s \to 0} s \cdot E(s)$$

By applying final value theorem.

$$= \lim_{s \to 0} s \cdot \frac{1}{1 + GD_{cl}} R$$

$$= \lim_{s \to 0} s \cdot \frac{1}{1 + GD_{cl}} \cdot \frac{1}{s^{k+1}}$$

We define GD_{cl} as follows

$$GD_{cl} = \frac{GD_{cl0}}{s^n}$$

what we are doing here is that we are separating out the number of poles at zero. For example:

$$GD_{cl} = \frac{(s+1)}{s(s+3)(s+2)}$$

Then, $$G D_{cl} = \frac{G D_{cl0}}{s}$$

where, $$G D_{cl0} = \frac{(s+1)}{(s+3)(s+2)}$$

Note that $G D_{cl0}$ does not have any pole at zero. With this modification,

$$e_{ss} = \lim_{s \to 0} s \cdot \frac{1}{1 + \frac{G D_{cl0}}{s^n}} \cdot \frac{1}{s^{k+1}}$$

$$= \lim_{s \to 0} s \cdot \frac{s^n}{s^n + G D_{cl0}} \cdot \frac{1}{s^{k+1}}$$

$$= \lim_{s \to 0} \frac{s^n}{s^n + G D_{cl0}} \cdot \frac{1}{s^k}$$

If $n = k = 0$, it is called type zero system.

$$e_{ss} = \frac{1}{1 + K_p}$$

K_p is the position constant.

$K_p = \lim_{s \to 0} G D_{cl}(s)$ [open loop transfer fun]

If $n = k = 1$, $e_{ss} = \frac{1}{K_v}$

type 1 system. K_v = velocity constant

$K_v = \lim_{s \to 0} s\, G D_{cl}(s)$ [open Loop]

It can track, step and ramp input

If $n = k = 2$, it is type 2 system.

$$e_{ss} = \frac{1}{K_a}$$

K_a = acceleration constant

$$K_a = \lim_{s \to 0} s^2 \, G D_{cl}(s) \text{ [open loop]}$$